

DATA WAREHOUSING AND DATA MINING

DAY - 2 LAB PROGRAMS

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Q1. Covariance and Correlation

Aim: To calculate the covariance and correlation between photograph preferences.

CODE:

```
A <- c(18,2,20)
B <- c(22,28,10)
C <- c(20,40,40)

cov(B,C)

data <- data.frame(A,B,C)

cov(data)

cor(B,C)

cor(data)
```

OUTPUT:

```
Console Terminal x Background Jobs x
R 4.5.2 ~ / ↗
> A <- c(18,2,20)
> B <- c(22,28,10)
> C <- c(20,40,40)
>
> cov(B,C)
[1] -20
>
> data <- data.frame(A,B,C)
> cov(data)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> cor(B,C)
[1] -0.1889822
>
> cor(data)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918 1.0000000 -0.1889822
C -0.4096440 -0.1889822 1.0000000
>
```

Q2. Equal-Frequency Partitioning & Data Smoothing

Aim: To perform equal-frequency partitioning, smoothing by bin mean and bin boundary, and plot a histogram.

CODE:

```
price <-  
c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,14,14,14,15,15,15,15,15,15,18,18,18,18,18,20,20,20,20,20,21,21,21,21,25,25,25,25,25,28,28,30)
```

```
bins <- 3
```

```
group <- cut(seq_along(price), breaks=bins, labels=FALSE)
```

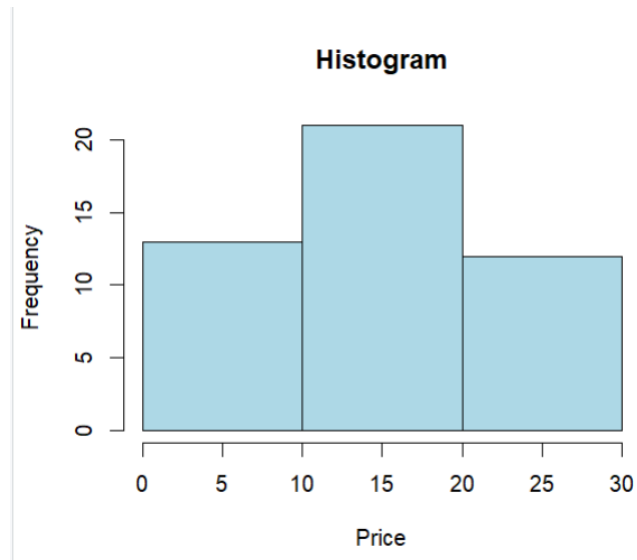
```
tapply(price, group, mean)
```

```
tapply(price, group, function(x) c(min(x), max(x)))
```

```
hist(price, breaks=bins, col="lightblue", main="Histogram", xlab="Price")
```

OUTPUT:

```
Console Terminal x Background Jobs x  
R R 4.5.2 ~/  
> price <- c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,14,14,14,15,15,15,15,15,15,18,18,18,18,18,20,20,20,20,20,21,21,21,21,25,25,25,25,25,28,28,30)  
>  
> bins <- 3  
>  
> group <- cut(seq_along(price), breaks=bins, labels=FALSE)  
>  
> tapply(price, group, mean)  
      1      2      3  
7.68750 16.93333 23.66667  
>  
> tapply(price, group, function(x) c(min(x), max(x)))  
$`1`  
[1]  1 14  
  
$`2`  
[1] 14 20  
  
$`3`  
[1] 20 30  
  
>  
> hist(price, breaks=bins, col="lightblue", main="Histogram", xlab="Price")
```



Q3. Mean, Median, Range & Boxplot

Aim: To compare the performance of two classes using mean, median, range, and boxplot.

CODE:

```
classA <- c(76,35,47,64,95,66,89,36,84)
```

```
classB <- c(51,56,84,60,59,70,63,66,50)
```

```
mean(classA)
```

```
median(classA)
```

```
range(classA)
```

```
mean(classB)
```

```
median(classB)
```

```
range(classB)
```

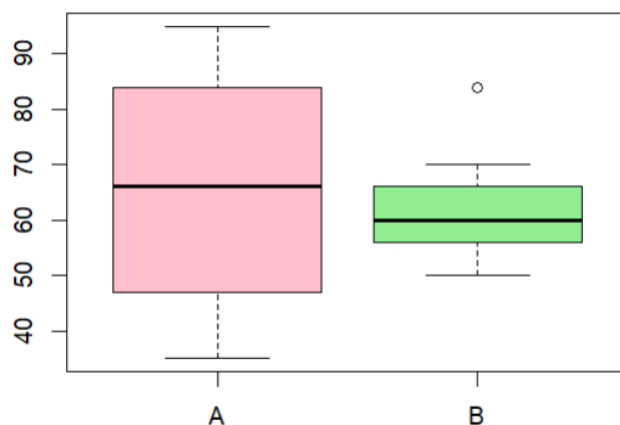
```
boxplot(classA,classB,
```

```
names=c("A","B"),
```

```
col=c("pink","lightgreen"))
```

OUTPUT:

```
Console Terminal x Background Jobs x
R 4.5.2 · ~/
> classA <- c(76,35,47,64,95,66,89,36,84)
> classB <- c(51,56,84,60,59,70,63,66,50)
>
> mean(classA)
[1] 65.77778
> median(classA)
[1] 66
> range(classA)
[1] 35 95
>
> mean(classB)
[1] 62.11111
> median(classB)
[1] 60
> range(classB)
[1] 50 84
>
> boxplot(classA,classB,
+         names=c("A","B"),
+         col=c("pink","lightgreen"))
> |
```



Q4. Min-Max & Z-Score Normalization

Aim: To perform Min-Max normalization and Z-score normalization.

CODE:

```
min <- 50000
```

```
max <- 100000
```

```
v <- 80000
```

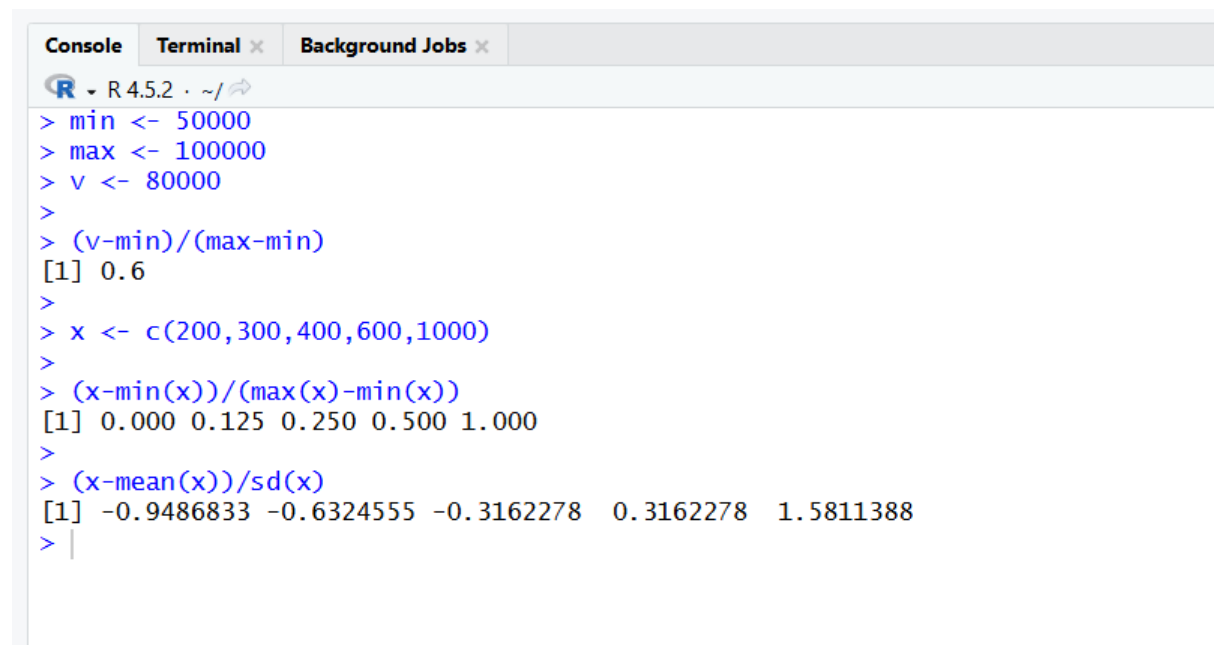
$(v - \min) / (\max - \min)$

`x <- c(200,300,400,600,1000)`

$(x - \min(x)) / (\max(x) - \min(x))$

$(x - \text{mean}(x)) / \text{sd}(x)$

OUTPUT:



```
R 4.5.2 · ~/
> min <- 50000
> max <- 100000
> v <- 80000
>
> (v-min)/(max-min)
[1] 0.6
>
> x <- c(200,300,400,600,1000)
>
> (x-min(x))/(max(x)-min(x))
[1] 0.000 0.125 0.250 0.500 1.000
>
> (x-mean(x))/sd(x)
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
> |
```

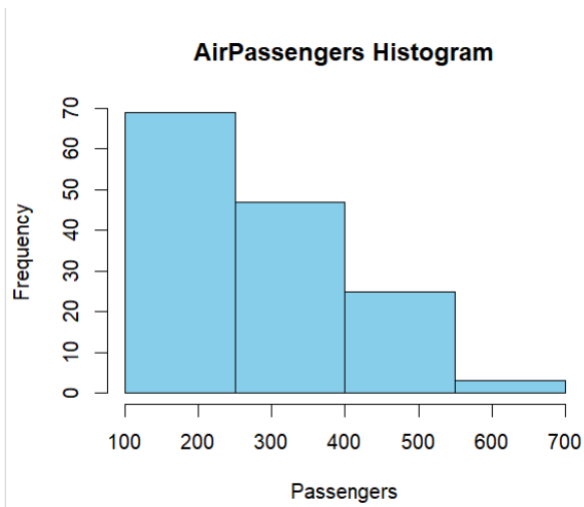
Q5. Histogram – AirPassengers

Aim: To create a histogram for the AirPassengers dataset.

CODE:

```
hist(AirPassengers,
     breaks=seq(100,700,150),
     col="skyblue",
     main="AirPassengers Histogram",
     xlab="Passengers")
```

OUTPUT:



Q6. Multiple Line Chart

Aim: To draw multiple line charts using mpg and qsec from mtcars.

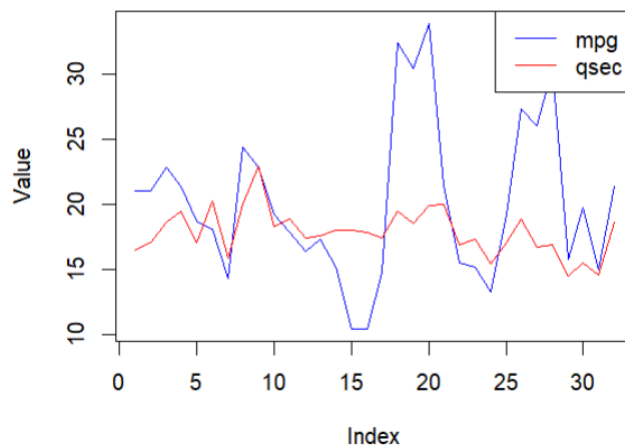
CODE:

```
plot(mtcars$mpg,  
     type="l",  
     col="blue",  
     ylim=range(c(mtcars$mpg,mtcars$qsec)),  
     xlab="Index",  
     ylab="Value")
```

```
lines(mtcars$qsec,col="red")
```

```
legend("topright",  
      legend=c("mpg","qsec"),  
      col=c("blue","red"),  
      lty=1)
```

OUTPUT:



Q7. Linear Regression

Aim: To find the relationship between hardness and mortality, fit a linear regression model, and predict mortality for hardness = 88.

CODE:

```
water <- data.frame(
  hardness=c(98,78,90,85,110,95,70,88,65,120),
  mortality=c(240,310,260,280,200,250,330,270,350,180)
)
```

```
plot(water$hardness, water$mortality,
     xlab="Hardness",
     ylab="Mortality",
     pch=19)
```

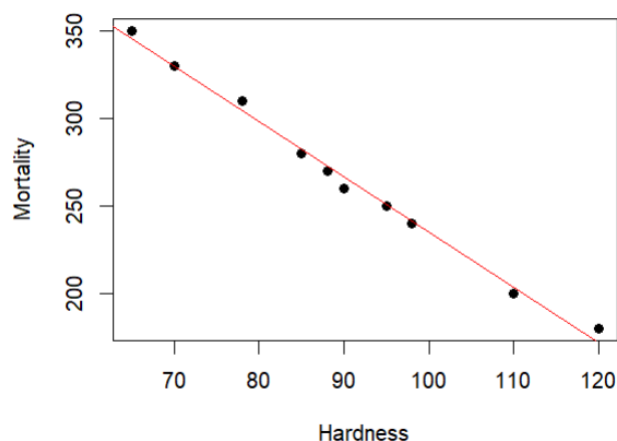
```
model <- lm(mortality ~ hardness, data=water)
```

```
abline(model,col="red")
```

```
predict(model,data.frame(hardness=88))
```

OUTPUT:

```
Console Terminal x Background Jobs x
R 4.5.2 ~ / ↗
> water <- data.frame(
+   hardness=c(98,78,90,85,110,95,70,88,65,120),
+   mortality=c(240,310,260,280,200,250,330,270,350,180)
+ )
> 
> plot(water$hardness, water$mortality,
+   xlab="Hardness",
+   ylab="Mortality",
+   pch=19)
> 
> model <- lm(mortality ~ hardness, data=water)
> 
> abline(model,col="red")
> 
> predict(model,data.frame(hardness=88))
1
273.0028
> |
```



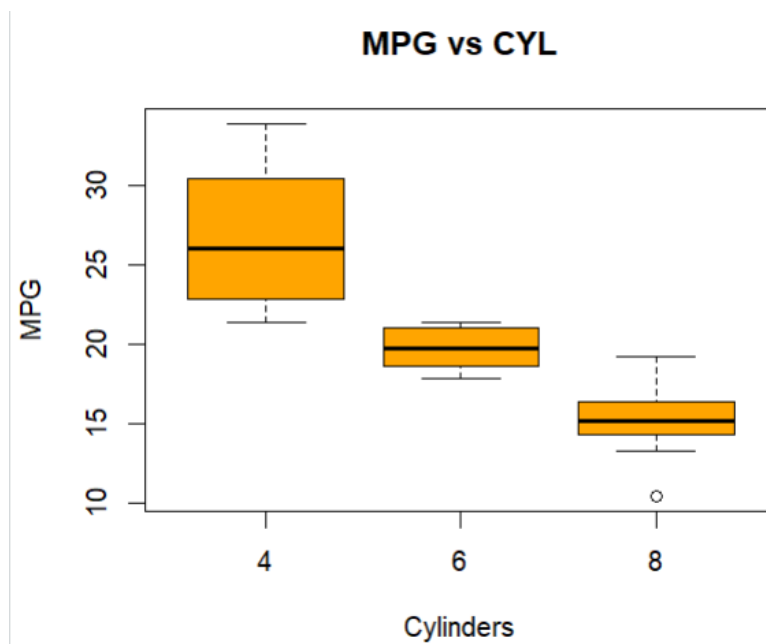
Q8. Boxplot (MPG vs CYL)

Aim: To create a boxplot showing the relationship between mpg and cyl.

CODE:

```
boxplot(mpg ~ cyl,
  data=mtcars,
  col="orange",
  xlab="Cylinders",
  ylab="MPG",
  main="MPG vs CYL")
```


OUTPUT:



Q9. Boxplot for Outlier Detection

Aim: To identify outliers in players' scores using a boxplot.

CODE:

```
score <- c(22,25,27,30,31,35,38,40,42,95)
```

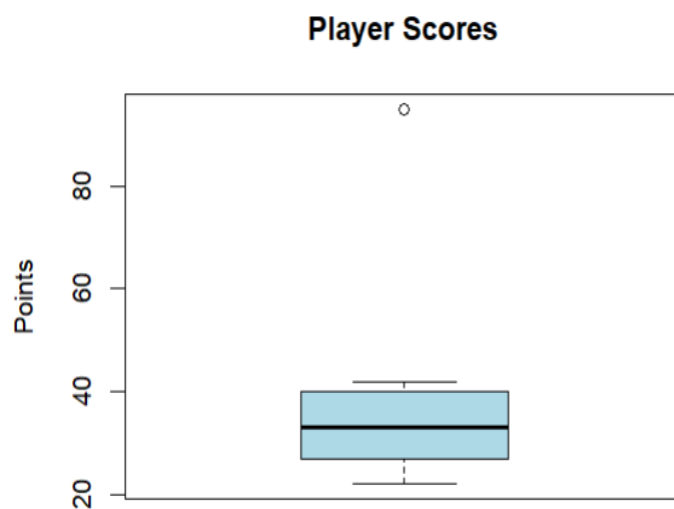
```
boxplot(score,
```

```
col="lightblue",
```

```
main="Player Scores",
```

```
ylab="Points")
```

OUTPUT:



Q10. Scatter Plot & Bar Chart (Blood Pressure vs Age)

Aim: To visualize the relationship between Age and Blood Pressure using scatter plot and bar chart.

CODE:

```
diabetes <- data.frame(  
  Age=c(25,30,35,40,45,50,55,60),  
  BloodPressure=c(70,72,75,80,82,85,88,90)  
)  
  
plot(diabetes$Age,  
     diabetes$BloodPressure,  
     pch=19,  
     col="blue",  
     xlab="Age",  
     ylab="Blood Pressure",  
     main="Age vs Blood Pressure")  
  
barplot(diabetes$BloodPressure,  
        names.arg=diabetes$Age,  
        col="green",  
        xlab="Age",  
        ylab="Blood Pressure")
```

OUTPUT:

